Chess 2 Impress – Automated Chess Board

Nathan MacPherson, Noah Beaty, Jack Tolleson, Allison Givens, Lewis Bates

Tennessee Technological University – Department of Electrical and Computer Engineering

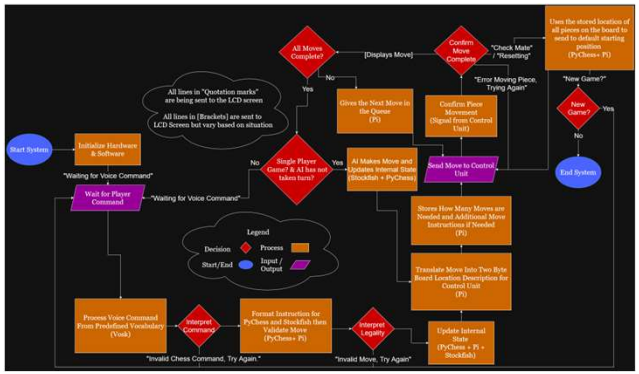
Abstract

Chess 2 Impress is an automated chess board that offers a new way to experience the game of chess. The goal of this project is to bring some of the advantages of playing chess digitally to people playing over the board. The system moves pieces with an electromagnet on a 2-axis pulley, takes voice commands from a microphone, and displays important information about the state of the game over an LCD screen.

Overview

Existing automated chess boards are impressive, but are inaccessible to most people, generally having high prices or sacrificing key features. One of the most important aspects of an automated chess board is being able to operate with only voice input. Hands-free input allows those with physical and visual disabilities to play while also bringing over the board and online chess one step closer to each other. To address these limitations, our team has designed an automated chess board that can move, capture, castle, and promote completely hands-free while remaining relatively cost effective. Alongside this, our board was constructed using affordable components to ensure it could be recreated by hobbyists looking to try an automated chess board.

Operational Flow Chart



Conceptual Design

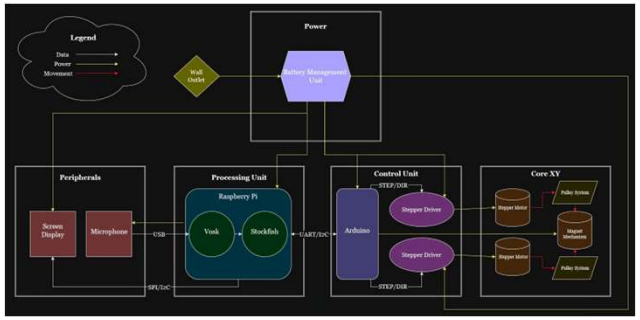
Our main chess algorithm is controlled by a Raspberry Pi 5, which listens for voice input using Vosk to process data from an MA5U Microphone and sends important game information to an LCD screen. The Pi uses the python chess library to validate moves and/or make its next move, then sends the move that will be played to an Arduino nano. The board uses a DFRobot BMS that powers our components with or without a wall plugin.

From Left to Right: Lewis Bates, Jack Tolleson, Nathan MacPherson, Allison Givens, Noah Beaty

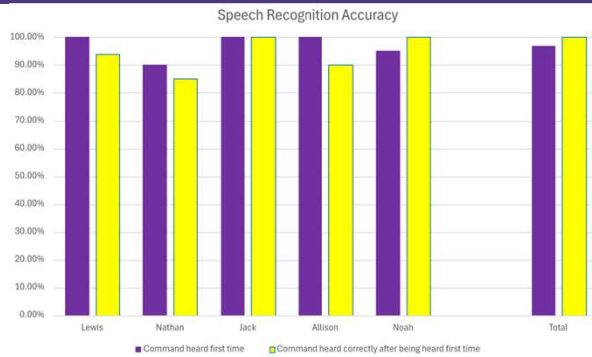
Conceptual Design

The Arduino nano takes move information from the Pi, then sends signals to our motors to execute the move. The pieces are moved by an electromagnet attached to a two-axis pulley system. Each piece has a neodymium magnet mounted in the base with a felt bottom to ensure the electromagnet can move them through a thin acrylic sheet.

Hardware Block Diagram



Testing



Voice recognition exceeds the 80% spec, with both first-try recognition (97.0%) and post-recognition correctness (95.0%) in the high 90s range across the five scored speakers. The small number of failures fall into two categories: three missed first-try recognitions, two incorrect move returns, and three commands using named-piece form that were misinterpreted as different moves. This suggests the pipeline is nearly perfect for coordinate-form commands but has slightly weaker disambiguation on named-piece commands for some speakers. Overall, the system passes the spec under these test conditions.

BOM

Subsystem	Price
Processing Unit	$135.02
Control Unit	$57.99
CoreXY Unit	$175.15
Peripherals Unit	$237.35
Power Unit	$185.39
Total	$721.24

Scan for full Bill of Materials:

Acknowledgements

Special thanks to our advisor Dr. Charles Van Neste, our capstone instructor Dr. Christopher Storm Johnson, Julie Mountain, Sara Vandermyde, Jeff Randolph, the Tennessee Tech Chess Club, and our respective families.